

# Synthesis of CuO and ZnO nanoparticles by a novel green route: Antimicrobial activity, cytotoxic effects and their synergism with ampicillin

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## ABSTRACT

CuO and ZnO nanoparticles (NPs) were synthesized at room-temperature following a novel, direct, high-yield (> 91%) and green method based on a mechanically activated metathesis reaction. The CuO-NPs exhibited a rod-like shape with average length and diameter of 22 and 6 nm, respectively, while the ZnO-NPs showed spherical-like morphology and mean diameter of 15 nm, as determined by TEM. The antimicrobial properties of the NPs and commercially available micrometric counterparts were evaluated by the micro dilutions method to determine the minimum inhibitory concentrations against reference strains from ATCC microorganisms, which were representative of the main groups of bacteria related to human infections, such as *E. coli*, *S. aureus*, *E. faecalis* and *P. aeruginosa*. The CuO and ZnO-NPs showed better antibacterial properties (500 µg/mL) in comparison with the commercial micrometric particles (1000 µg/mL). The analysis of the synergism between both NPs and ampicillin showed a decrease of their concentrations against all the microorganisms studied, with the ZnO-NPs showing better synergism activity than CuO-NPs, whereas the combination of the antibiotic with ZnO-NPs showed a six-fold decrease against all the microorganisms studied except to *P. aeruginosa*. The FIC index was calculated with both NPs and ampicillin to each bacterium. In most cases, the effect was partial synergism, and an additive effect was achieved when those combinations were evaluated against *P. aeruginosa*. The cell viability tests showed that the CuO and ZnO-NPs synthesized and mixed with the antibiotic are biocompatible in effective doses (< 25 µg/mL). The NPs used in this study, prepared by a green innovative synthesis, and their combination with a basic antibiotic showed interesting results, suggesting the possibility of one alternative to biological applications.

## 1. Introduction

Since the first half of the last century, when Alexander Fleming

discovered penicillin, antibiotics have been routinely used for the treatment of bacterial infections. Its use has been an important factor to increase life expectancy and reduce mortality due to infectious diseases

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in vulnerable populations such as children or elderly people. Currently, there is excessive use of antimicrobials, and in many cases, this is unjustified, which has resulted in the existence of microorganisms that have evolved to develop resistance to various antibiotics, representing a major health problem [1].

The number of people dying from bacterial infections that do not respond to traditional treatments with antibiotics is increasing. The numbers indicate around 25,000 dead people a year only in Europe, which results in an increase in the costs of health services by almost 1500 million euros per year [2].

In addition to the resistance that bacteria can develop, it must also be considered that they have the ability to organize themselves into biofilms, ecosystems of one or several species with complex functional and structural characteristics, which potentiates their capacities for growth, survival and virulence. Given these developed intrinsic and adaptive resistance mechanisms, the efficacy of antibiotics has been significantly reduced [3]. In this context, various alternatives have been proposed, among which the following can be mentioned: inhibition of quorum sensing, phage therapy, inhibition of lecithin, iron chelation, vaccination strategies, antimicrobial peptides, electrochemical scaffolds, nanoparticles, as well as the combination of some of these alternatives in search of synergies [3–6].

The use of NPs is a viable alternative to compensate the growing resistance of microorganisms to common antibiotics due to their size, a feature that allows them to alter the formation of biofilms and easily penetrate the membrane of microorganisms [7]. Particularly, with the development of the synthesis of NPs via the green chemistry routes, attention has been directed to their use taking advantage of this approach [8]. NPs with antibacterial properties based on metallic and metallic/nonmetallic elements (oxides) have been investigated. Among these, perhaps, the most common are: Ag, Au, TiO<sub>2</sub>, ZnO, CuO, FeO and Al<sub>2</sub>O<sub>3</sub> [9]. Some of them present adequate biocompatibility which is attractive for antimicrobial applications in humans [10]. The mechanisms by which the NPs are able to interfere with the viability of microorganisms are multiple [11] and shall not be discussed here.

It is possible to prepare NPs of different chemical composition, crystalline structure, morphology, specific surface area, surface functional chemical groups, etc., and to take advantage of their substantially improved properties, among them, particularly, the antimicrobial activity can be enhanced [12]. NPs can be synthesized by several methods, such as: thermal reduction, vapor deposition, microwave irradiation, sonochemical, synthesis of metal vapors, chemical reduction, among many other methods [13,14]. In recent years, green chemistry procedures have attracted significant scientific interest, because they are safe and environmentally friendly methods due to the reduction in the use of reagents and that toxic byproducts are not produced [15]. These methods attempt to maximize the efficiency in the synthesis of NPs and to reduce production costs considerably. There are three important aspects to be taken into account in this type of methodologies: the reaction medium, the reducing agent and the stabilizing substances for NPs, which must be friendly to the environment [16].

NPs based on various metal oxides have been considered for their broad spectrum of antimicrobial activity and lower propensity to induce resistance [17]. Other characteristics that recommend the use of NPs based on oxides are related to their chemical and thermal stability, low density, they can be prepared by various simple, synthetic routes, and without the use of strong reducing agents, preparation at room-temperature, lower cost, possibility of reconditioning and reusing them in some technological applications, etc. [12].

Examples of these NPs are zinc and copper oxides (ZnO, CuO), which have been previously studied and shown to be highly effective against American Type Culture Collection (ATCC) strains such as *Escherichia coli* and *Staphylococcus aureus* in planktonic form [18]. The activity of these NPs depends on their size, morphology, concentration, dispersion and their stability in the medium [14,19]. The mechanisms by which these NPs generally act are: i) the accumulation of NPs inside

the microorganism, ii) the formation of reactive oxygen species, iii) release of ions in growth media and iv) electrostatic interactions [11].

The combination of therapies in the search for synergistic effects could constitute an important advance against microbial resistance. These synergisms are explained by adding the mechanisms that present NPs to traditional antibiotics that show low effectiveness and promoting the formation of new “nanobioantibiotics” that could be an effective way to combat the problem of microbial resistance [19].

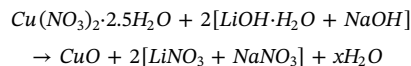
Previous combinations of NPs and antibiotics have already been tested. Nishanthi et al. (2019) used combinations of NPs, synthesized by green methods, of silver, gold and platinum with a variety of commercial antibiotics against *Bacillus* sp. This streptomycin-resistant microorganism became highly susceptible to the combination of gold-streptomycin NPs [20]. López-Carrizales et al. (2018) found partial and additive synergisms of the silver NPs in combination with ampicillin and amikacin in clinical strains resistant to multiple drugs obtained from patients with urinary tract infections associated with catheters [21]. The previously evaluated NPs have presented promising results that suggest continuing the investigation of nanobioantibiotics. However, it must be taken into account that these novel NPs of metals, despite being synthesized by green methods, are still expensive due to the precursor materials and that their synthetic methods can be complicated and time-consuming. Nowadays, scientific reports of synergism studies using cost-effective oxide-type NPs with antibiotics against the most representative microorganisms related with human illnesses, are scarce. On the other hand, copper and zinc oxides are inorganic materials with well-known antimicrobial properties. Further, Cu and Zn are considered as strategic inorganic materials and they are two of the main exported metals in Mexico.

The main goal of this article is to report the novel, room-temperature, green synthesis of CuO and ZnO-NPs by following a direct, economical, efficient, high-yield and one-step method. The synthesized NPs showed antimicrobial properties improving their bioactivity by generating synergies with ampicillin.

## 2. Materials and methods

### 2.1. Synthesis of CuO and ZnO-NPs

Cu(NO<sub>3</sub>)<sub>2</sub>·2.5H<sub>2</sub>O, LiOH and NaOH (Merck, ACS reagent, ≥ 98%) were used as starting reagents for the preparation of CuO-NPs. The reagents were weighed in the appropriate amounts and deposited in a YSZ mortar (Yttria Stabilized Zirconia) along with the grinding medium (YSZ balls of 12.7 mm in diameter), to be subsequently deposited in a SPEX mill (8000 M) during 45 min (3 cycles of 15 min) at a speed of 1725 rpm. The ratio of reagents/balls used was 1/10. The amount of reagents used were balanced so that there was no Cu(NO<sub>3</sub>)<sub>2</sub>, LiOH or NaOH excess (1:2 M ratio) according to the following equation that describes the overall process:



where  $x$  corresponds to the number of moles of H<sub>2</sub>O and the molar ratio of LiOH/NaOH is equal to 0.54/0.46.

The milled precursor was transferred to a Teflon box and subjected to a drying process at 80 °C overnight. Subsequently, several washes were carried out using deionized water and ethanol to remove the soluble nitrates, whereas the water-insoluble NPs were recovered by centrifugation and dried for 2 h at 80 °C.

The synthesis of ZnO-NPs was previously reported by Mendoza-Mendoza et al. (2018) following a similar one-step procedure [22].

### 2.2. Characterization of NPs

The milled precursor and CuO-NPs were characterized by X-ray

diffraction (XRD) using a Rigaku Ultima IV diffractometer operated at 35 kV and 25 mA with CuK $\alpha$  radiation ( $\lambda = 1.541 \text{ \AA}$ ). The XRD patterns were recorded from 10 to 80° (2 $\theta$ ) at the scan rate of 0.02°/s. The identification of the crystalline phases was carried out by comparison with crystallographic standard data reported in the International Center of Diffraction Data (ICDD). The crystallite size (CS) was obtained by the Halder-Wagner method implemented in the PDXL2 software of the diffractometer. The thermal analysis of the milled precursor was realized by differential scanning calorimetry/thermogravimetric analysis (DSC/TGA) in a Setaram Labsys Evo 1600 thermoanalyzer, using approximately 5 mg of the sample which was heated from room temperature to 550 °C at a rate of 10 °C/min. The morphology and size of CuO and ZnO-NPs were observed by transmission electron microscopy (TEM) in a FEI TITAN 80–300 kV microscope. High resolution transmission electron microscopy (HRTEM) and selected area electron diffraction (SAED) techniques were used to observe structural characteristics of the as-synthesized NPs. In addition, CuO and ZnO commercial powders supplied by Merck (ACS reagent,  $\geq 99\%$ ) were characterized by XRD to obtain their structural characteristics for comparative purposes.

### 2.3. Conservation and reactivation of the strains used in this study

Reference strains, representative of the main bacterial groups *E. coli* ATCC 25922, *Pseudomonas aeruginosa* ATCC 27853, *S. aureus* ATCC 25923 and *Enterococcus faecalis* ATCC 29212, from the ATCC, were used, kept in vials with glycerol suspension and then stored at  $-20^\circ\text{C}$ . The reactivation of the strains consisted in the gradual thawing of each of the vials. Once the contents of the vials were homogenized, 200  $\mu\text{L}$  representative of each vial was taken and mixed in a vial containing 1000  $\mu\text{L}$  of Mueller Hinton broth, which was incubated 24 h at a temperature of 37 °C in aerobic conditions.

### 2.4. Standardization of antimicrobial tests

The antimicrobial activity was evaluated by the microdilution method in broth to determinate the minimum inhibitory concentration (MIC), this method was validated using reference strains of *S. aureus* ATCC 25923 and *E. coli* ATCC 25922. The commercial antibiotics used were oxacillin (Sigma-Aldrich, Germany), 64  $\mu\text{g/mL}$ , and ampicillin (Sigma-Aldrich, Germany), 128  $\mu\text{g/mL}$ , respectively. The end points were determined when no turbidity in the well was observed and corroborated the cut points according to the Clinical and Laboratory Standards Institute (CLSI). The evaluations were carried out in triplicate on three different days [23].

### 2.5. Evaluation of the antimicrobial activity of NPs

The antimicrobial activity of the NPs and their commercially available micrometric homologous counterparts was evaluated on two Gram negative microorganisms from the *E. coli* ATCC 25922 and *P. aeruginosa* ATCC 27853, as well as, two Gram positive microorganisms: *S. aureus* ATCC 25923 and *E. faecalis* 29212. The NPs, as well as, their commercially available micrometric homologous counterparts were diluted in water to a concentration of 1000  $\mu\text{g/mL}$ ; serial dilutions were carried out in the range of 1000 to 0.9  $\mu\text{g/mL}$ . Then, a bacterial suspension, adjusted to an absorbance of 0.08 at 600 nm (approximately,  $1.5 \times 10^8$  UFC/mL), was diluted 1:1000 and added to each well. Microplates were incubated 24 h at 37 °C in aerobic conditions in a humid chamber. The MIC was determined in the cut points where the turbidity was a minimum. To corroborate the cut points, subcultures on agar plate were carried out and determined the minimal bactericidal concentration. The averages of the MIC and their standard deviations were obtained from three independent tests carried out on different days of each of the NPs with representative strains of Gram positive and Gram negative bacteria.

### 2.6. Synergy tests

The antimicrobial activity of the NPs, as well as of the antibiotic alone (ampicillin 128  $\mu\text{g/mL}$ ) or in combination was determined by serial dilutions in 96-well plates in accordance with the guidelines of the CLSI. Two fold dilutions of each, CuO and ZnO-NPs, and the antibiotic were distributed into the wells of the plate to get the designed range of concentrations. The dilutions performed on the NPs started from the columns to the right and the antibiotic dilutions were started from the first row downwards, each of the wells held a unique combination of concentrations of the NPs and the antibiotic. Once the antimicrobial wells of known concentration are titrated and those MICs exerted the expected effect in the presence of NPs with the antibiotic and vice versa are determined, the fractional inhibitory concentration (FIC) of each component was determined based on the individual MICs. According to Guelfi et al. (2013) the synergistic effect of two antimicrobial substances can be classified through the FIC as follows: synergistic effect ( $\Sigma \text{FIC} \leq 0.5$ ), partial synergistic effect ( $\Sigma \text{FIC} > 0.5$  and  $< 1$ ), additive effect ( $\Sigma \text{FIC} = 1$ ), indifferent effect ( $\Sigma \text{FIC} > 1$  and  $\leq 4$  and antagonistic effect ( $\Sigma \text{FIC} > 4$ ) [24].

### 2.7. Cell viability assay

To analyze if CuO and ZnO-NPs affect cell viability, calcein/etidium homodimer (EthD-1) assay was performed in human dermal fibroblasts. Dermal fibroblasts were isolated from donated skin from aesthetic surgeries, previous signature of a consent letter by the patient. Cells were cultured with DMEM/F12 medium (Gibco) supplemented with 10% of fetal bovine serum (Gibco) and 1% of penicillin/streptomycin (Gibco). 20,000 fibroblasts were seeded in 48 well culture dishes. After 24 h of adhesion the treatments were applied: 5, 10, 15, 20, 25, 30, 35 and 40 ppm of ZnO or CuO-NPs, besides the control condition (PBS). After 24 h of the treatments live/dead viability/cytotoxicity kits were used (Thermo), 1  $\mu\text{M}$  calcein AM and 2  $\mu\text{M}$  etidium homodimer (EthD-1) were added to the culture medium and kept in contact during 1 h; then the medium was changed and photographs were taken using an epifluorescence microscope (Zeiss, AvioObserver). The number and percentage of cells were calculated using Image J software. The assays were performed by triplicate and at three independent times.

## 3. Results and discussion

### 3.1. Synthesis and characterization of NPs

Fig. 1 (a) shows the XRD pattern obtained after milling the initial reagents mixture for the preparation of CuO-NPs. The characteristic reflections of NaNO<sub>3</sub> (JCPDS 85–0850) and CuO (JCPDS 80–1268) crystalline phases were identified, whose standards reported in the ICDD have been included in the bottom of Fig. 1 to facilitate comparison. The presence of the alkaline nitrates confirmed the occurrence of the metathesis reaction in the milling process. During the metathesis reaction, the Li<sup>+</sup> and Na<sup>+</sup> cations displace Cu<sup>++</sup> ion to react with the NO<sub>3</sub><sup>−</sup> anions and producing NaNO<sub>3</sub> and LiNO<sub>3</sub>. The detection of CuO confirms the crystallization of the desired phase in the milling medium itself, which also corroborates the novel production of CuO at room-temperature by mechanosynthesis. Fig. 1 (b) contains the DSC/TGA curves of the milled precursor. In the DSC curve, the existence of an endothermic peak located at 201 °C is revealed, which is related to the congruent fusion of the eutectic composition LiNO<sub>3</sub>/NaNO<sub>3</sub> (0.54/0.46 M ratio) [25]. Although the LiNO<sub>3</sub> phase was not detected by XRD, the aforementioned thermal event demonstrates its presence in the reaction system and also constitutes additional evidence of the metathesis reaction experienced in the system. Likewise, there are no other thermal events related to the existence of non-reacted LiOH or NaOH in milling, which must be represented by endothermic peaks close to 462 and 318 °C, correspondingly. The TGA curve shows a weight loss of

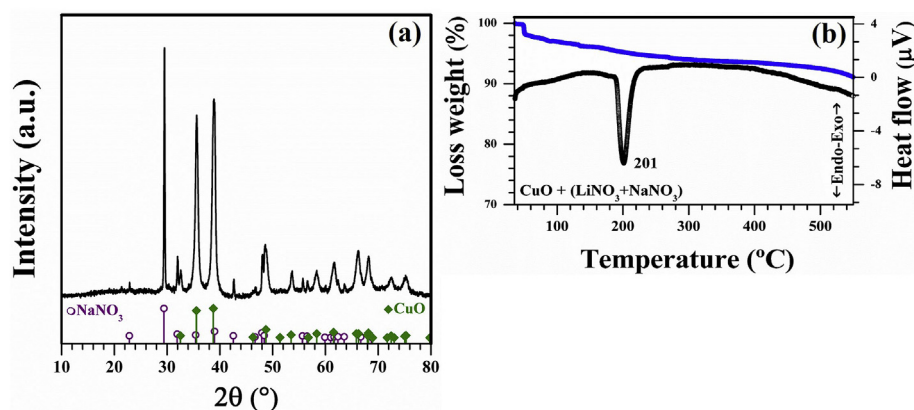


Fig. 1. XRD pattern (a) and DSC/TGA curves (b) of the milled precursor for preparing CuO-NPs.

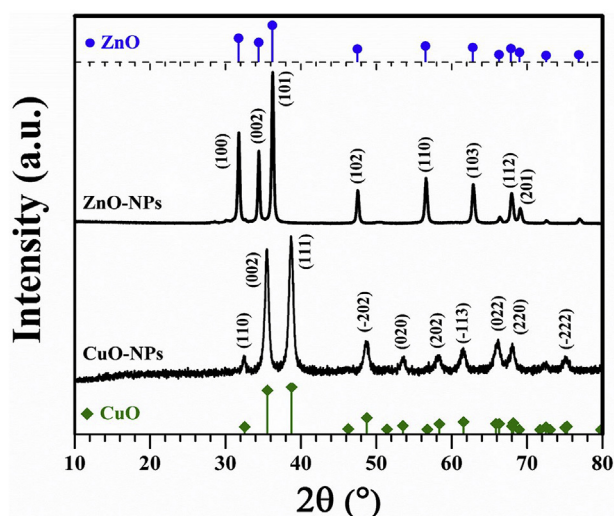


Fig. 2. XRD patterns of CuO and ZnO-NPs.

approximately 10% from 35 to 550 °C, which demonstrates the thermal stability of alkali nitrates in this temperature interval.

Fig. 2 shows the XRD patterns of CuO and ZnO-NPs. The patterns are constituted by the characteristic reflections of the structures with monoclinic and hexagonal symmetry of CuO (JCPDS 80–1268) and ZnO-NPs (JCPDS 80–0075), respectively. Also, it can be observed the absence of additional peaks related to undesirable crystalline species.

Table 1 shows the data of cell volume, crystallite size (CS) and average characteristic dimensions of the particles obtained for commercial and as-synthesized CuO and ZnO-NPs. The CS determined by the Halder-Wagner method reported values of 29.8 and 58.1 nm for commercial particles of CuO and ZnO, respectively, whereas, 14.5 and 10.7 nm were the sizes obtained for the CuO and ZnO-NPs,

Table 1

Cell volume, crystallite size (CS) and particle size distribution (PSD) of the samples under study.

| Sample               | Cell volumen (Å <sup>3</sup> ) | CS (nm) | Average characteristic dimensions (nm)                |
|----------------------|--------------------------------|---------|-------------------------------------------------------|
| CuO-NPs <sup>a</sup> | 81.19                          | 14.5    | 22 (length), Std. Dev. 9<br>6 (diameter), Std. Dev. 2 |
| CuO-Com <sup>a</sup> | 81.17                          | 29.8    | –                                                     |
| ZnO-NPs <sup>b</sup> | 47.72                          | 10.7    | 15 (diameter), Std. Dev. 3                            |
| ZnO-Com <sup>b</sup> | 47.60                          | 58.1    | –                                                     |

<sup>a</sup> ICDD, JCPDS 80-1917.

<sup>b</sup> ICDD, JCPDS 80-0075.

correspondingly.

Fig. 3 (a) shows a TEM image of the CuO-NPs. It can be observed that the particles showed rod shapes and heterogeneous nanometric sizes. The morphology and size of the particles can be seen in greater detail in Fig. 3 (b), corresponding to the region selected with the blue dashed line in part (a). Fig. 3 (c) presents an image obtained by HRTEM, which reveals the high crystallinity of the synthesized particles that were prepared at room-temperature. The crystallinity in long-range ordered arrays is manifested by the observation of fringes of atoms that correspond to the atomic positions of the crystalline planes of the material's structure. The measurement of the spacing between the fringes was 0.25 nm, which is related to the (002) plane of the CuO structure. In Fig. 3 (d) it is presented the SAED image obtained for the set of particles contained in the delimited area with the red dashed line in Fig. 3 (a). In this, the main crystalline reflections of the monoclinic structure of CuO were indexed to the (110), (002), (111), (–202), and (–113) planes. Fig. 3 (e) shows the TEM image of ZnO-NPs and it is observed particles with spherical-like shapes and nanometric sizes that are forming agglomerates. Moreover, histograms for CuO and ZnO-NPs obtained by measuring particle size distribution from TEM images are included in Fig. 3. The histograms showed the unimodal distribution for particles of CuO-NPs with main diameter and length of 22 and 6 nm, respectively, whereas 15 nm of diameter for ZnO-NPs. These dimensions are also reported in Table 1. Herein, the main particle sizes are in agreement with the CS determined by XRD.

### 3.2. Antimicrobial activity

After realizing the standardization with the ATCC strains and the commercial antibiotics (Supplementary Table 1), the antimicrobial activity of the NPs synthesized by a novel green method was evaluated. The NPs showed antibacterial properties at concentrations of 500 μg/mL against all the microorganisms studied except to *P. aeruginosa* ATCC 27853. In comparison, their commercially available counterparts showed activity at higher concentrations, 1000 μg/mL (Table 2). In general, the commercial particles at the concentrations studied showed no antimicrobial activity. This difference in antimicrobial behavior can probably be explained because smaller materials show better antimicrobial properties. Hoseinzadeh et al. (2017) describe that NPs smaller than 30 nm seem to be the most effective because they present a greater surface of interaction and in turn can reach the interior of bacteria more easily [12]. In the same sense, for the CuO-NPs, Azam et al. (2012) showed that the smaller size of the NPs increases the surface area and that their concentration, therefore, also increases the effects that the CuO-NPs could have on bacteria [26]. Since the size is important for NPs to show antimicrobial activity, here a standard method was carried out in a nutrient medium, which had influence in the size of the NPs. In our experiments the NPs showed an increased size



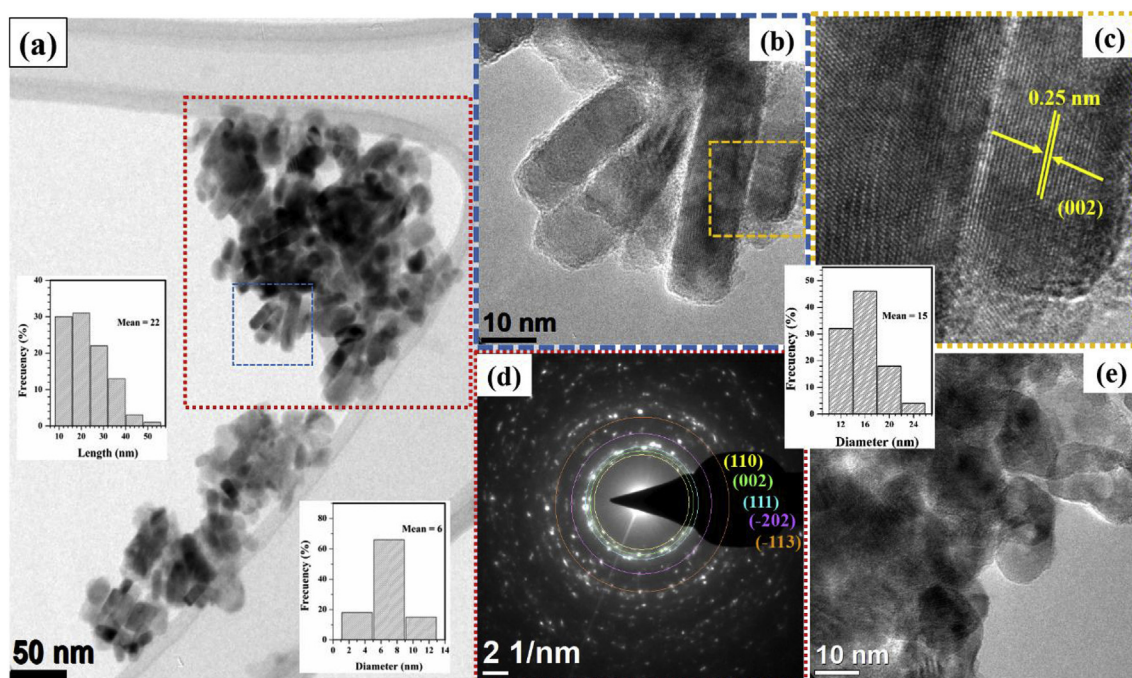


Fig. 3. TEM (a and b), HRTEM (c) and SAED images of the as-prepared CuO-NPs. TEM (e) image of ZnO NPs (e). The histograms show the particle size distribution of CuO-NPs (a) and ZnO-NPs (e).

due to their agglomeration in the culture medium, reaching a range between 800 - 1000 nm for both CuO and ZnO-NPs. Those results are in accordance with previous reports, which indicate the increase of the NPs size when they are diluted in nutrient medium containing organic molecules as is the case of the culture medium used here [27]. Other authors showed different results of the MIC when the conditions of the experimentation were changed [28,29].

The exact antimicrobial mechanism that these NPs follow has not yet been identified; however, it is likely that the main mechanism of the ZnO-NPs is the generation of hydrogen peroxide, in addition to the accumulation of NPs on the surface of the bacteria, the generation of reactive oxygen species (ROS) and the liberation of Zn ions [30]. In the CuO-NPs the main mechanism may be different, the passage of the NPs through the bacterial cell membrane and the damage to the interior of the bacterium may be the way in which these CuO-NPs are toxic for the bacteria [16]. Even though the antimicrobial activity of the NPs against *P. aeruginosa* was minor than for the other microorganisms, it could be explained for the biochemical activity of the *Pseudomonas* genus.

### 3.3. Synergistic activity tests between NPs and ampicillin

The analysis of the synergism between both NPs and ampicillin showed a decrease of their concentrations to inhibit all the

microorganisms studied, the combination of ampicillin with ZnO-NPs showed better antimicrobial activity than the combination of ampicillin with CuO-NPs (Table 3), the combination of the antibiotic with ZnO-NPs showed six-fold decrease for the strains of *E. coli* ATCC 25922, *S. aureus* ATCC 25923 and *E. faecalis* ATCC 29212, whereas that for *P. aeruginosa* ATCC 27853 a three-fold antimicrobial reduction activity was observed. The antimicrobial activity of the CuO-NPs was better against *E. faecalis* ATCC 29212 with a six-fold reduction, while that of *P. aeruginosa* ATCC 27853 showed just a one-fold reduction. The synergistic effects of both NPs and ampicillin were evaluated for each bacterium. The FIC was calculated (Table 4), in most cases the effect was partial synergism. Interestingly, additive effects results were achieved when those combinations were evaluated against *P. aeruginosa*. This behavior can be attributed to the possible stability that the presence of ampicillin gives to ZnO-NPs. Previous works have shown that this combination allows to have a better dispersion, which favors the contact area and with it the antimicrobial activity [21].

The decrease in the concentration of NPs in order to achieve antimicrobial effects is vital, since the toxicity that these NPs could present, depends to a large extent on concentration [31], as well as on the secondary components that accompany the reaction. These elements are significantly diminished when the NPs are synthesized by green methods. The effects of synergism between NPs and antibiotics have

Table 2

MIC tests of the green synthesis NPs and commercial ZnO and CuO on ATCC strains.

| Oxide   | <i>E. coli</i> 25922 |     | <i>S. aureus</i> 25923 |     | <i>P. aeruginosa</i> 27853 |     | <i>E. faecalis</i> 29212 |     |
|---------|----------------------|-----|------------------------|-----|----------------------------|-----|--------------------------|-----|
|         | MIC µg/mL            |     | MIC µg/mL              |     | MIC µg/mL                  |     | MIC µg/mL                |     |
|         | $\bar{X}$            | SD  | $\bar{X}$              | SD  | $\bar{X}$                  | SD  | $\bar{X}$                | SD  |
| ZnO NPs | 500                  | ± 0 | 500                    | ± 0 | 1000                       | ± 0 | 500                      | ± 0 |
| CuO NPs | 500                  | ± 0 | 500                    | ± 0 | 500                        | ± 0 | 500                      | ± 0 |
| ZnO c   | 1000                 | ± 0 | 500                    | ± 0 | 1000                       | ± 0 | 1000                     | ± 0 |
| CuOc    | 1000                 | ± 0 | 1000                   | ± 0 | 1000                       | ± 0 | 1000                     | ± 0 |

ZnO NPs, Zinc Oxide nanoparticle; CuO NPs, Copper Oxide nanoparticle; ZnOc, commercial micrometric Zinc Oxide; CuOc, commercial micrometric Copper Oxide; ATCC, American Type Culture Collection.

**Table 3**  
MIC tests of green ZnO and CuO synthesis NPs combined with ampicillin on ATCC strains.

| NPs              | <i>E. coli</i> 25922 |       | <i>S. aureus</i> 25923 |       | <i>P. aeruginosa</i> 27853 |       | <i>E. faecalis</i> 29212 |       |
|------------------|----------------------|-------|------------------------|-------|----------------------------|-------|--------------------------|-------|
|                  | MIC µg/mL            |       | MIC µg/mL              |       | MIC µg/mL                  |       | MIC µg/mL                |       |
|                  | $\bar{X}$            | SD    | $\bar{X}$              | SD    | $\bar{X}$                  | SD    | $\bar{X}$                | SD    |
| <b>ZnO + Amp</b> | 6.47                 | ± 2.3 | 7.81                   | ± 0.0 | 125                        | ± 0.0 | 7.81                     | ± 0.0 |
| <b>CuO + Amp</b> | 15.62                | ± 0.0 | 15.62                  | ± 0.0 | 250                        | ± 0.0 | 7.81                     | ± 0.0 |

ZnO NPs, Zinc Oxide nanoparticle; CuO NPs, Copper Oxide nanoparticle; Amp, Ampicillin; ATCC, American Type Culture Collection.

been previously tested. Nishanthi et al. (2018) obtained good results when combining NPs of silver, gold and platinum with various antibiotics, suggesting this methodology as a possible alternative for future treatment [20]. This antimicrobial activity is a remarkable result; however, standard methods are necessary to replicate the biological conditions, as well as, the cytotoxicity evaluation is relevant to determine potential applications.

The exact mechanism of how the combination of CuO and ZnO-NPs works with ampicillin has not been fully studied; however, it is likely that the damage caused to the bacterial wall plays an important role. This approach is similar to that established by Bhande et al. (2013), who observed a delay in the growth curve of Gram negative microorganisms [31]. In fact, these results of antimicrobial activity of ZnO-NPs in combination with beta-lactam antibiotics are in accordance with the results analyzed here. In both research works, the combination of the ZnO-NPs with the antibiotics evaluated showed poor inhibition against *P. aeruginosa*; however, other authors explain that ZnO-NPs can inhibit the virulence factors and biofilm formation of *P. aeruginosa* without affecting cell growth [32]. This finding suggests an opportunity area, since *P. aeruginosa* is a common microorganism with strong relation to human infection by biofilm formation on medical devices [33].

### 3.4. Cytotoxic effect of CuO and ZnO-NPs

To analyze the cytotoxic effect of CuO-NPs on dermal fibroblasts a calcein/EthD-1 assay was performed. The concentrations of 5 and 10 ppm of CuO-NPs did not affect cell viability but 15, 20, 25, 30, 35 and 40 ppm showed significant differences compared with the control (Fig. 4). Even though in all treatments, the percentage of viability is high (73.8% in the 40 ppm condition), the number of viable cells decreases almost twice in the highest concentration (1.9 times less cells in 40 ppm compared with the control). This suggests that when the concentrations of NPs increase, many of the dead cells were detached from the culture plate, but most of the cells that remained attached were viable. While other CuO-NPs, in a range of 20–40 nm, showed greater cytotoxic effects where viability was reduced by 90% at 20 ppm in human lung epithelial cell A549, our study reveals a less cytotoxic effect on epithelial fibroblasts placed in contact with CuO-NPs synthesized by the green route reported here where viability was reduced around 20% at 20 ppm [34]. In the case of ZnO-NPs all the treatments showed almost 100% of cell viability, only the 35 and 40 ppm

concentrations showed significant differences compared with the control condition. However, the number of cells decreased in a modest way from 10 to 35 ppm. Only for 40 ppm the reduction is 1.6 times less than the control condition (Fig. 5). Previous results showed cytotoxic effect with commercial ZnO-NPs in a range of 50 to 20 nm, which presented a cytotoxic effect on the viability of neural cells above 10 ppm, similar with the results obtained in this study. In addition, it is reported that the main mechanism of cytotoxicity is the generation of reactive oxygen species when the nanoparticles come into contact with acid medium corresponding to lysosomes [35]. These results suggest that most of the dead cells were not attached to the culture dishes, and the cells that remain in the cultures showed high cell viability.

## 4. Conclusions

CuO and ZnO-NPs were successfully synthesized following an innovative, direct, efficient, room-temperature, high-yield and green method, which is based on a metathesis reaction activated by mechanical milling. CuO-NPs were prepared using the by-product eutectic mixture of LiNO<sub>3</sub>/NaNO<sub>3</sub> as inorganic media which has no precedents. The CuO and ZnO-NPs suffered a direct crystallization in the milling media, which is called mechanosynthesis. This route represents an alternative for the easy and large-scale production of nanostructured particles and, also, it can be extended to the preparation of other antibacterial inorganic agents. The synthesized CuO-NPs showed rod-like morphologies with length and average diameters of 22 and 6 nm, respectively, whereas the ZnO-NPs presented spherical-like shapes with average diameters of 15 nm. Both the CuO and ZnO-NPs offer antimicrobial effects by themselves and these effects are favored when adding the commonly used antibiotic ampicillin; at the concentrations used in this research the effect they have on eukaryotic cells have adequate toxicity, which should continue being exploited with the intensification of the functionalization of these nanoparticles for possible biological applications. More studies are necessary to evaluate the behavior of this possible alternative against a panel of more diverse microorganisms to understand the mechanism of action.

## Acknowledgments

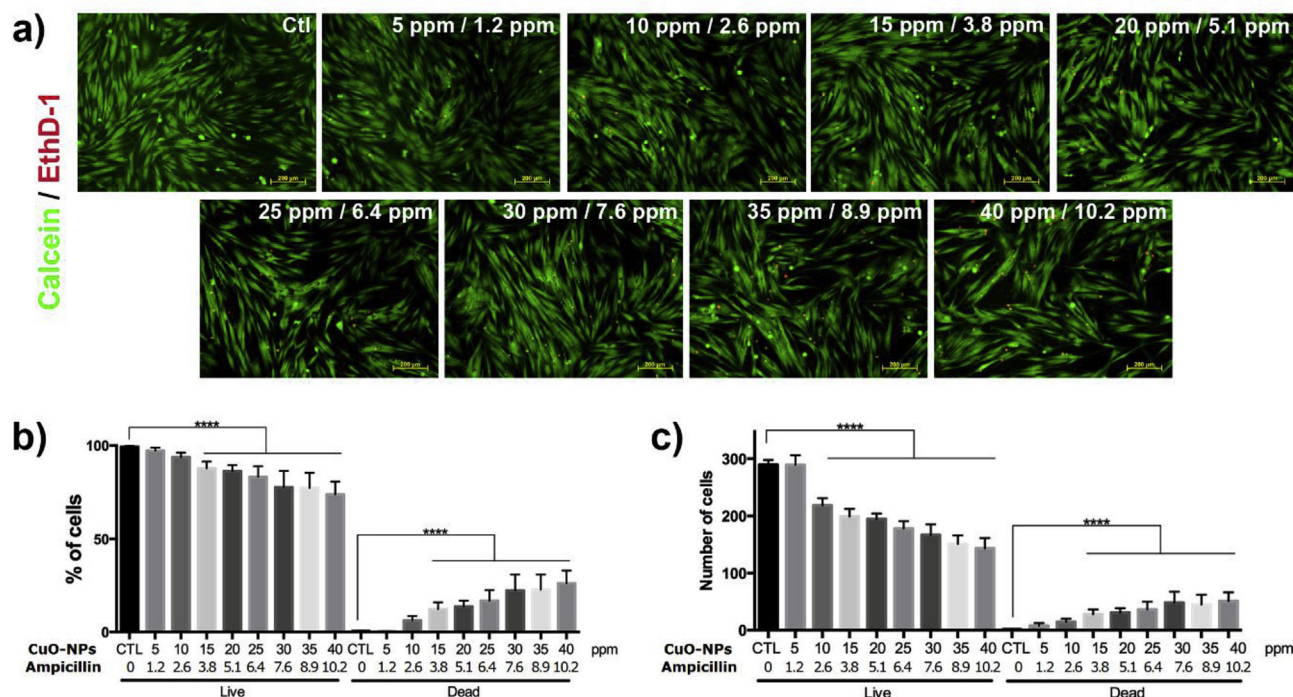
Financial support for this work was partially provided by Consejo Nacional de Ciencia y Tecnología (CONACYT), Mexico through project

**Table 4**  
FIC obtained from the combination of NPs and ampicillin on ATCC strains.

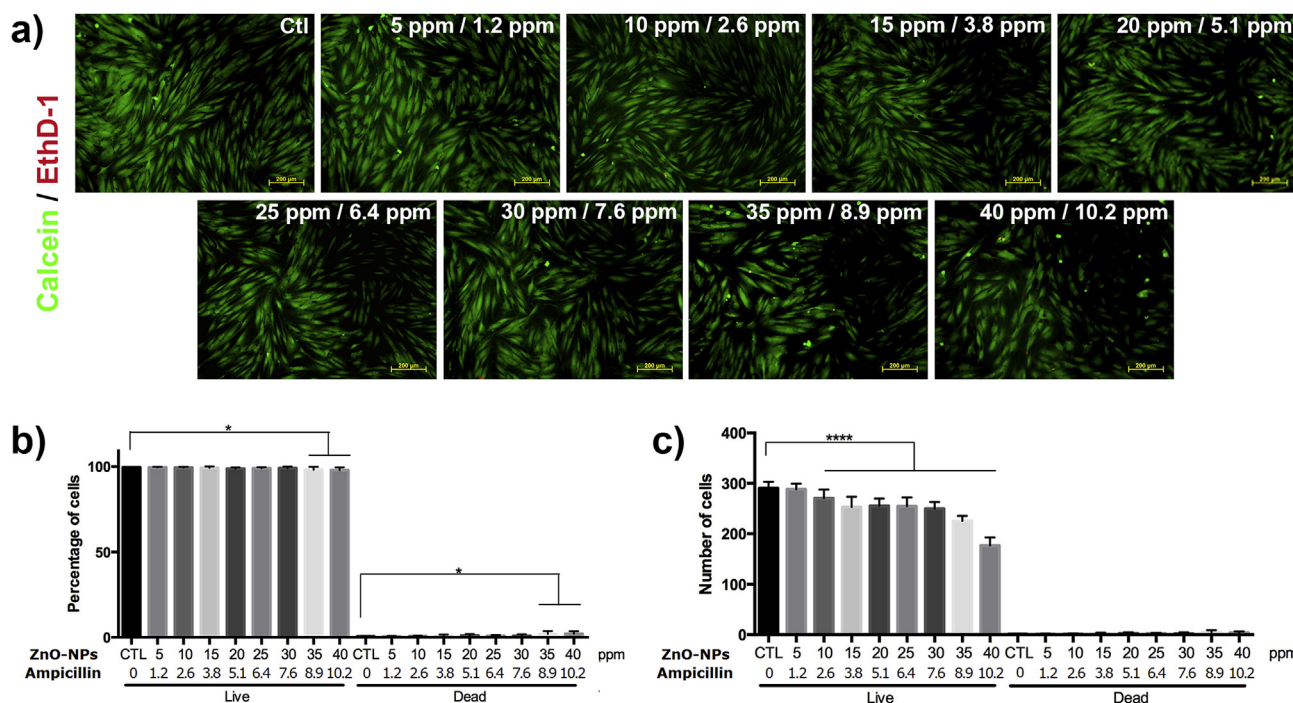
| NPs              | <i>E. coli</i> 25922 |        | <i>S. aureus</i> 25923 |        | <i>P. aeruginosa</i> 27853 |        | <i>E. faecalis</i> 29212 |        |
|------------------|----------------------|--------|------------------------|--------|----------------------------|--------|--------------------------|--------|
|                  | MIC µg/mL            |        | MIC µg/mL              |        | MIC µg/mL                  |        | MIC µg/mL                |        |
|                  | $\bar{X}$            | SD     | $\bar{X}$              | SD     | $\bar{X}$                  | SD     | $\bar{X}$                | SD     |
| <b>ZnO + Amp</b> | 0.99                 | ± 0.00 | 0.99                   | ± 0.00 | 1.03                       | ± 0.88 | 0.99                     | ± 0.03 |
| <b>CuO + Amp</b> | 0.99                 | ± 0.01 | 0.99                   | ± 0.01 | 1.06                       | ± 0.08 | 0.99                     | ± 0.53 |

FIC, Fractional Inhibitory Concentration; ZnO NPs, Zinc Oxide nanoparticle; CuO NPs, Copper Oxide nanoparticle; Amp, Ampicillin; ATCC, American Type Culture Collection.





**Fig. 4.** Effect of CuO-NPs on cell viability. a) The images show viable cells in green (Calcein positives) and dead cells in red (EthD-1 positives) with the different treatments. b) The graph shows the percentage of cell viability at the different experimental treatments. c) The graph shows the number of cells live and dead with the different treatments. \*\*\*\*p < 0.0001 one-way ANOVA by running a Tukey's post hoc test. Scale bars correspond to 200 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)



**Fig. 5.** Effect of ZnO-NPs on cell viability percentages. a) The images show viable cells in green (calcein positives) and dead cells in red (EthD-1 positives) in the different treatments. b) The graph shows the percentage of viable and dead cells with the different treatments. c) The graph shows the number of viable cells and dead cell with the different treatments. \*p < 0.01, \*\*\*\*p < 0.0001 one-way ANOVA by running a Tukey's post hoc test. Scale bars correspond to 200 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ceramint.2019.08.171>.

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